
Time-Adaptive Masked DiT with Minibatch Optimal Transport Pairing for Class-Conditional Image Generation

Zhixuan Jiang
Carnegie Mellon University
Pittsburgh, PA 15213
zhixuanj@andrew.cmu.edu

Mahika Gunjkar
Carnegie Mellon University
Pittsburgh, PA 15213
mgunjkar@andrew.cmu.edu

Mahir Nagersheth
Carnegie Mellon University
Pittsburgh, PA 15213
mnagersh@andrew.cmu.edu

Abstract

Diffusion Transformers achieve strong class-conditional image generation, but they remain expensive to train and sample. We study whether two training-time interventions—minibatch optimal-transport coupling for flow matching and time-adaptive token masking for latent DiTs—interact constructively or merely provide independent gains. We implement both mechanisms in a class-conditional latent flow Transformer and evaluate them with a controlled 2×2 factorial design at fixed DiT-XL scale. Under a standardized Heun-40, CFG-1.75 offline FID_{5k} protocol, neither OT coupling nor adaptive masking improves FID when applied alone, while their composition achieves the best sample quality. The resulting interaction is non-additive: the combined cell outperforms the additive prediction from the two isolated interventions by 1.9139 FID points. Because the present evidence is single-seed, uses a 5,000-sample FID protocol, and includes a mask-only batch-size caveat, we interpret the result as evidence for a constructive interaction in this implementation rather than as a fully resolved effect size. These results suggest that coupling design and token visibility should be considered jointly when training efficient flow-based Transformers.

1 Overview and Context

Diffusion models [Ho et al., 2020, Song et al., 2021] and their continuous-time flow-matching counterparts [Lipman et al., 2023, Liu et al., 2023] have become widely used for high-fidelity class-conditional image synthesis. The Diffusion Transformer (DiT) [Peebles and Xie, 2023] demonstrates that a Transformer backbone scales competitively with U-Net designs, and when paired with a latent encoder [Rombach et al., 2022] enables high-resolution synthesis at manageable compute. Nonetheless, training DiT at scale remains computationally expensive, even on reduced-resolution class-conditional benchmarks.

Two distinct training-accelerator strategies have emerged in the recent literature. The first improves the coupling between noise and data samples: OT-CFM [Tong et al., 2024] and Multisample Flow Matching [Pooladian et al., 2023] replace the independent noise–data sampling of standard flow matching with a minibatch-level optimal transport assignment, yielding straighter integration trajectories and reducing the number of function evaluations required at inference. The second reduces per-step compute: MaskDiT [Zheng et al., 2024] and the Masked Diffusion Transformer [Gao et al., 2023] randomly drop patch tokens during training, letting the model process shorter sequences per gradient step with no inference-time cost.

1.1 Motivation and Objectives

These two accelerators act on different axes: OT pairing changes the noise–data coupling in the flow-matching target, whereas adaptive masking changes the token sequence processed inside the DiT forward pass. We ask whether they help independently or interact non-additively when combined.

This final stage builds on the midterm system, which validated the ImageNet-100 DDPM/DDIM training and sampling pipeline and identified latent diffusion and classifier-free guidance as planned extensions. The midterm samples also showed that early pixel-space DDPM training remained in a structured-noise regime, motivating the move to latent space. We completed the latent/class-conditional infrastructure but replaced the original DDPM-vs-DDIM speedup target with a latent flow-matching DiT and Euler/Heun ODE sampling. Our objectives are: **O1**, reproduce a fixed asymmetric latent DiT-XL flow-matching baseline; **O2**, measure isolated OT-CFM and adaptive-masking effects; and **O3**, quantify the OT $\times$ mask interaction via a fixed-scale 2×2 factorial at a 400-epoch horizon.

2 Related Work and Background

2.1 Literature Review

Flow Matching and Rectified Flow. Flow matching [Lipman et al., 2023] and Rectified Flow [Liu et al., 2023] recast continuous-time diffusion as the regression of a conditional velocity field along straight interpolation paths between noise and data. Under linear interpolation between a noise endpoint z_0 and a data endpoint z_1 , the per-sample velocity target is the closed-form constant $v^*(x_t, t) = z_1 - z_0$, making the training objective a simple regression on straight-line paths.

Optimal Transport in Flow Matching. Tong et al. [2024] and Pooladian et al. [2023] observe that the marginal flow-matching objective is agnostic to how noise and data are paired within a minibatch, and that optimizing this pairing via minibatch-level Wasserstein-2 assignment produces straighter trajectories and lower-variance gradients. We adopt the minibatch Hungarian variant with an L^2 transport cost; a Sinkhorn relaxation is available in principle for larger batches.

Masked Training for Diffusion Transformers. Zheng et al. [2024] and Gao et al. [2023] show that randomly dropping patch tokens during training accelerates convergence on compute-normalized budgets without degrading inference quality, since full token sequences are still processed at sample time. We use a time-adaptive variant that drops more tokens near the noise-dominated end, where masking is expected to be less damaging than near the data endpoint, and tapers to no masking at the data end.

Diffusion Transformers and Latent Diffusion. Peebles and Xie [2023] replaces the U-Net backbone of DDPM [Ho et al., 2020] with a Transformer operating on image patches. Latent Diffusion Models [Rombach et al., 2022] perform diffusion in a VAE-compressed latent space, reducing spatial resolution by an order of magnitude and making high-resolution synthesis practical. Our method combines both: DiT in VAE latent space at 32×32 latent resolution (from 128×128 pixels).

2.2 Background

Evaluation metric. We use Fréchet Inception Distance (FID) [Heusel et al., 2017] as the primary quality metric. FID computes the squared Fréchet (Wasserstein-2) distance between Gaussian fits of ImageNet-pretrained Inception-V3 features from generated and reference images. It captures both sample fidelity and class-level diversity, and is also the standard metric reported by prior DiT and MaskDiT work [Peebles and Xie, 2023, Zheng et al., 2024], providing a common reference point subject to the protocol differences discussed below.

Baseline architecture. Our baseline is an asymmetric latent DiT-XL flow-matching model trained without OT pairing and without adaptive masking. We choose DiT-XL because it is the highest-compute regime in the Peebles–Xie scale family, where training-time improvements are most consequential. Keeping the baseline within one backbone family isolates training-regime changes (OT, masking) from architecture-scale differences.

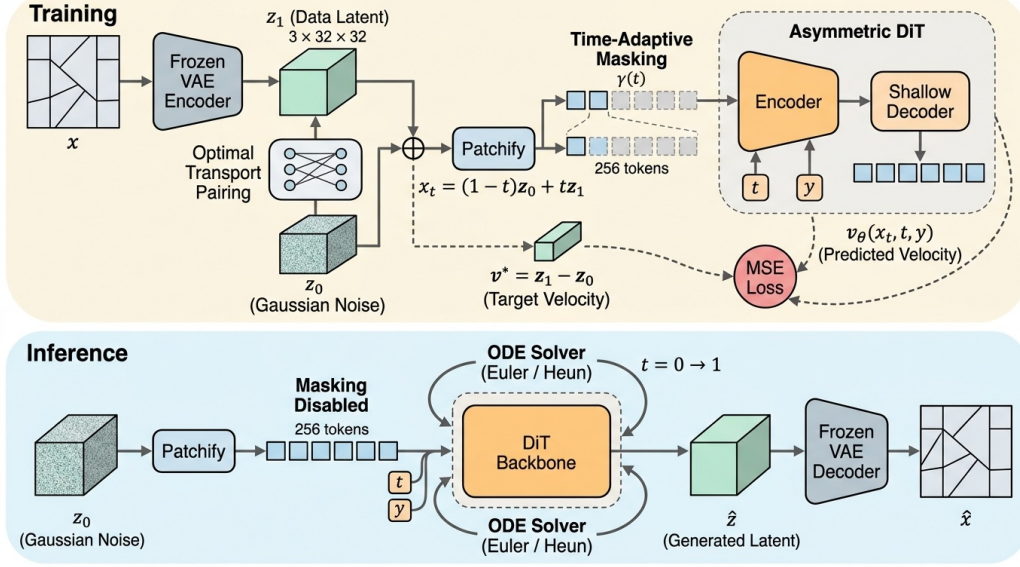


Figure 1: System overview of OT-AdaMask Flow. During training, an input image is encoded into a VAE latent z_1 , paired with Gaussian noise z_0 through minibatch OT, interpolated to $x_t = (1-t)z_0 + tz_1$, and passed through an asymmetric DiT whose encoder operates on a time-adaptively masked token sequence while the decoder reconstructs the full latent grid. The regression target is the constant velocity $v^* = z_1 - z_0$. During inference, masking is disabled, the learned ODE is integrated from $t = 0$ to $t = 1$ with Euler or Heun steps, and the final latent is decoded with the frozen VAE.

Dataset choice. We train on ImageNet-100 [Deng et al., 2009], a standard 100-class subset of ImageNet. At 128×128 resolution, full ImageNet-1k exceeds our compute budget, while ImageNet-100 retains class-conditional diversity and is widely used as a controlled intermediate benchmark in diffusion-model studies.

3 Methodology

We train a class-conditional velocity-prediction model $v_\theta(x_t, t, y)$ using the flow-matching objective with two modifications: minibatch OT pairing in the training data pipeline, and time-adaptive random token masking inside the DiT forward pass. Throughout this section, B_{local} denotes per-GPU micro-batch size (the batch that actually enters one model forward), and $B_{\text{global}} = B_{\text{local}} \times N_{\text{GPU}}$ denotes effective global batch size under DDP. Figure 1 summarizes the exact training and inference pipelines implemented in the repository.

3.1 Model Description

3.1.1 Flow Matching Objective

Let $x \in \mathbb{R}^{3 \times 128 \times 128}$ denote an input image and let $z_1 = s_{\text{vae}} \text{Enc}_{\text{vae}}(x)$ be its scaled latent representation, with $s_{\text{vae}} = 0.1845$ and $z_1 \in \mathbb{R}^{3 \times 32 \times 32}$. We sample $z_0 \sim \mathcal{N}(0, I)$ with the same shape and form the linear interpolant

$$x_t = (1-t)z_0 + tz_1, \quad t \sim \mathcal{U}[0, 1].$$

The velocity target is $v^* = z_1 - z_0$, constant along the straight path. At inference, we integrate the learned ODE $\dot{x}_t = v_\theta(x_t, t, y)$ forward from $t = 0$ to $t = 1$. For a step size $\Delta t = 1/N_{\text{FE}}$, the two solvers used in the codebase are

$$x_{t+\Delta t}^{\text{Euler}} = x_t + \Delta t v_\theta(x_t, t, y)$$

and the second-order Heun update

$$x_{t+\Delta t}^{\text{Heun}} = x_t + \frac{\Delta t}{2} (v_1 + v_2),$$

with

$$v_1 = v_\theta(x_t, t, y), \quad v_2 = v_\theta(x_t + \Delta t v_1, t + \Delta t, y).$$

The scheduler implements Euler directly and Heun through a second model evaluation in the inference pipeline.

3.1.2 Minibatch Optimal Transport Pairing

Rather than drawing (z_0, z_1) independently per sample, we first sample a local minibatch $\{(z_0^{(i)}, z_1^{(i)})\}_{i=1}^{B_{\text{local}}}$ on each GPU and then permute the noise batch via a minibatch optimal assignment

$$\pi^* = \arg \min_{\pi \in S_{B_{\text{local}}}} \sum_{i=1}^{B_{\text{local}}} \|z_1^{(i)} - z_0^{(\pi(i))}\|_2^2,$$

where the cost is computed on flattened latent vectors in the same scaled VAE coordinate system used by the regression loss. The assignment is solved exactly with the Hungarian algorithm in $O(B_{\text{local}}^3)$ time. For minibatches exceeding $B_{\text{local}} = 64$, our implementation raises an explicit error and recommends a Sinkhorn relaxation; all runs in this report keep $B_{\text{local}} \leq 64$, so exact Hungarian matching is used throughout. Under distributed training, OT pairing is applied independently on each GPU’s local batch, with no cross-device coupling. The resulting permutation preserves the marginal distributions of z_0 and z_1 while replacing the independent coupling with a lower-cost joint pairing. Prior work shows that such couplings can reduce path curvature and gradient variance in flow matching [Tong et al., 2024, Pooladian et al., 2023]. Our DiT-XL factorial below tests whether that coupling remains beneficial after adding latent DiT training, classifier-free guidance, and adaptive token masking.

3.1.3 Time-Adaptive Patch Masking

During training only, we randomly drop a fraction of the DiT patch tokens after patchification and positional embedding. The mask ratio is a function of the flow time $t \in [0, 1]$:

$$\gamma(t) = \gamma_{\max} \cdot \cos^2\left(\frac{\pi t}{2}\right), \quad \gamma_{\max} = 0.75.$$

This schedule drops the maximal fraction at $t = 0$ (the noise-dominated end, where we heuristically expect prediction to rely more on global context than fine local token detail) and tapers to zero at $t = 1$ (data, where fidelity matters most). At inference the mask ratio is clamped to zero: the full token sequence is always processed by the model, so masking introduces no inference-time token dropping or sequence-length reduction.

Let $N = (32/2)^2 = 256$ be the latent patch-token count for 128×128 images with VAE latent size 32 and patch size 2. At each step, the encoder sees

$$N_{\text{keep}}(t) = \max\left(1, \text{round}((1 - \gamma(t))N)\right).$$

Since training samples $t \sim \mathcal{U}[0, 1]$, the expected keep ratio under the ideal schedule is

$$\mathbb{E}[N_{\text{keep}}/N] \approx 1 - \mathbb{E}[\gamma(t)] = 1 - \gamma_{\max} \int_0^1 \cos^2\left(\frac{\pi t}{2}\right) dt = 1 - \frac{\gamma_{\max}}{2} = 0.625.$$

In the implementation, a single scalar mask ratio is computed from the first sample’s t in each minibatch (an unbiased estimator of the expected batch-level ratio under i.i.d. time sampling) to avoid synchronization overhead. Each sample first receives an i.i.d. random score for every patch token; the model keeps the $N_{\text{keep}}(t)$ lowest-scoring tokens, processes only those tokens in the deep encoder, and then restores the full sequence by re-inserting learned mask tokens before the shallow decoder. Attention FLOPs in encoder blocks scale approximately as $O(N_{\text{keep}}^2)$, so this schedule reduces the dominant training-time attention cost while preserving a full-resolution decoder output for the regression loss.

Scale	Hidden	Depth _{enc}	Depth _{dec}	Heads	Params (M)
DiT-S	384	12	2	6	37.79
DiT-B	768	12	2	12	150.79
DiT-L	1024	24	2	16	494.57
DiT-XL	1152	28	2	16	721.47

Table 1: DiT backbone scales used in this work. Hidden size, encoder depth, and head count follow Peebles and Xie [2023]. An additional 2-block decoder is appended to each scale following the asymmetric MaskDiT design of Zheng et al. [2024]. “Params (M)” are trainable parameters of the DiT backbone only (excluding the class embedder, which adds 0.04–0.12M depending on scale). DiT-B is listed here for configuration-space completeness but is *not* trained in the preliminary sweep of §5.1 due to compute-budget constraints; its row is retained to make the four-scale configuration explicit and to facilitate future extensions.

3.1.4 DiT Backbone in VAE Latent Space

We operate in VAE latent space: input images at 128×128 are encoded to 32×32 latent maps (3 latent channels in this implementation) by a pretrained VAE, then patchified with patch size 2 into 256 tokens. The DiT backbone uses adaLN conditioning on the flow time t and the class label y , and is instantiated at four scales — DiT-S, DiT-B, DiT-L, DiT-XL — matching the hidden-size/depth/head configurations of Peebles and Xie [2023]. We additionally use an asymmetric encoder/decoder split following Zheng et al. [2024]: the decoder depth is set to 2 transformer blocks on top of the standard encoder depth. All transformer blocks use AdaLN-Zero conditioning, and the final adaptive layer and output projection are zero-initialized as in the DiT implementation recipe, which stabilizes early training. We note that the midterm plan assumed a common $8 \times$, 4-channel latent layout; the final implementation uses the repository’s actual $3 \times 32 \times 32$ latent tensor for 128×128 inputs, and all shape equations below follow this code path. We emphasize that this 2-block asymmetric decoder is applied *uniformly* across all four cells of the 2×2 ablation (§5.2) and across all scales of the preliminary sweep (§5.1), so it does not confound the OT $\times$ mask attribution *within* our factorial design. It does, however, constitute an architectural deviation from the pure DiT-XL backbone of Peebles and Xie [2023] and from the original MaskDiT decoder configuration; we therefore treat it as a known confound for any *cross-paper* comparison to those works, and note that isolating the independent contribution of the 2-block decoder would require an additional baseline not included in the present submission. Table 1 summarizes the four DiT scales with implementation-exact trainable parameter counts computed from the actual code configuration.

3.1.5 Class Conditioning and Classifier-Free Guidance

Class labels are embedded through a learned lookup table with $100+1$ entries: the first 100 correspond to ImageNet-100 classes, and the final slot is a dedicated null token. During training, each label is replaced by the null token with probability 0.1, following classifier-free guidance [Ho and Salimans, 2022]. At inference, the flow pipeline evaluates the model on both the conditional and null-token batches and combines the predicted velocities as

$$v_{\text{cfg}}(x_t, t, y) = v_\theta(x_t, t, y_\emptyset) + w \left(v_\theta(x_t, t, y) - v_\theta(x_t, t, y_\emptyset) \right),$$

where w is the guidance scale. The standardized offline evaluations and qualitative panel in this report use Heun-40 sampling with $w = 1.75$. We fix this inference-time hyperparameter across all four DiT-XL cells so that the FID comparison isolates the training-side variables — OT pairing, adaptive masking, and their interaction — rather than being confounded by per-model guidance tuning.

3.2 Dataset

All experiments use ImageNet-100 [Deng et al., 2009], a 100-class subset of ImageNet, resized to 128×128 resolution. Online augmentation is limited to random horizontal flipping. Latents are produced on-the-fly during training by a frozen VAE encoder under `torch.no_grad()` and then scaled by the fixed VAE factor of 0.1845 before being passed to the denoiser. Batch sampling follows the standard shuffled DataLoader pipeline (or DistributedSampler under DDP), i.e., without-replacement sampling within each epoch and `drop_last=True` for shape consistency across devices.

The final DiT-XL factorial comparison is reported at global batch size 128; the preliminary scale sweep uses experiment-dependent per-GPU batches chosen to fit each model scale. For offline class-conditional evaluation, the inference script generates 50 samples per class, yielding exactly 5,000 images for each checkpoint comparison.

3.3 Evaluation Metric

We use Fréchet Inception Distance (FID) [Heusel et al., 2017] as the primary quality metric. Given two sets of images X_r (reference) and X_g (generated), their Inception-V3 pool-3 features are extracted and fit to multivariate Gaussians $\mathcal{N}(\mu_r, \Sigma_r)$ and $\mathcal{N}(\mu_g, \Sigma_g)$. The FID is the squared Fréchet distance between these two Gaussians:

$$\text{FID}(X_r, X_g) = \|\mu_r - \mu_g\|_2^2 + \text{tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r^{1/2}\Sigma_g\Sigma_r^{1/2})^{1/2}\right),$$

where $\Sigma_r^{1/2}$ and the outer square root denote the unique symmetric positive semi-definite matrix square roots. Lower values indicate closer agreement between the two distributions in Inception-V3 feature space, subsuming both sample fidelity and inter-class diversity in a single scalar. We distinguish two evaluation paths in the codebase. First, an *optional* online FID hook in `train.py` can periodically sample `fid_num_samples` images (default: 2048) against a cached reference subset; however, all ablation YAMLS reported here set `fid_every_n_epochs=0`, so this online hook is disabled. Second, the *offline* evaluation path invokes `inference.py` to generate exactly 5,000 images (50 per class) and `generate_submission.py` to package their Inception statistics into the course submission CSV. When a fixed reference statistics file is provided, the same offline path also computes local FID from those 5,000 generated images. Whenever we refer to FID_{5k} in tables, we mean this local 5,000-sample FID protocol, not the external leaderboard score. Although `inference.py` also logs Inception Score, the final ranking claim uses FID_{5k} to match the controlled factorial protocol. This budget follows the course-specified evaluation workflow, but it is below the 50,000-sample convention adopted by many published DiT and MaskDiT results, so any cross-paper comparison should be read with that sample-count caveat in mind.

3.4 Loss Function

The training objective is the squared- L^2 flow-matching regression loss, computed on each paired example after the OT assignment of §3.1.2. Let $\tilde{z}_0^{(i)} = z_0^{(\pi^*(i))}$ denote the noise latent paired with data latent $z_1^{(i)}$. Then

$$x_t^{(i)} = (1-t)\tilde{z}_0^{(i)} + tz_1^{(i)}, \quad u^{(i)} = z_1^{(i)} - \tilde{z}_0^{(i)}, \\ \mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,i} \left[\|v_\theta(x_t^{(i)}, t, y^{(i)}) - u^{(i)}\|_2^2 \right].$$

When OT pairing is disabled, $\tilde{z}_0^{(i)}$ is simply the originally sampled noise paired with $z_1^{(i)}$. The interpolant is the same flow-matching path defined in §3.1.1, and v_θ is the DiT with the masking rule of §3.1.3 applied during the forward pass. Although the encoder processes only visible tokens, the decoder reconstructs the full token grid and the loss is computed on the full latent velocity map $(B, 3, 32, 32)$.

Optimization details. We optimize θ with AdamW [Loshchilov and Hutter, 2019] at learning rate $\eta_0 = 3 \times 10^{-4}$ (DiT variants) or $\eta_0 = 1 \times 10^{-4}$ (U-Net reference), weight decay 1×10^{-4} , and gradient clipping at global L^2 norm 1.0. The learning-rate schedule is cosine annealing (no warmup) to $\eta_{\min} = 10^{-6}$:

$$\eta(s) = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos \left(\pi \cdot \frac{s}{S_{\max}} \right) \right),$$

for $s \in [0, S_{\max}]$. All runs use bfloat16 mixed-precision training under PyTorch DDP. We optimize the denoiser and class embedder jointly, maintain an exponential moving average of denoiser weights with decay 0.9999, and use EMA weights for epoch-end sampling and checkpoint evaluation. Classifier-free guidance dropout is applied during training by randomly replacing the class label y with a null token with probability 0.1; the standardized offline evaluation uses CFG scale 1.75.

3.5 Experimental Depth and Iteration

Design-space exploration proceeded in four phases, each with committed YAML configs in `configs/ablations/`. Phase 1 configured a scale sweep with both OT and adaptive masking enabled: exp01 (DiT-XL), exp02 (DiT-L), exp03 (DiT-B, configured but not included as a completed sweep run), and exp04 (DiT-S). The completed progress table reports only runs with usable W&B summaries. Phase 2 probed the masking mechanism at DiT-B scale: exp05 used a constant 50% mask ratio, exp06 used time-adaptive masking with $\gamma_{\max} = 0.5$, exp07 used $\gamma_{\max} = 0.9$, exp08 removed the asymmetric decoder (`dit_decoder_depth=0`), and exp09 disabled masking entirely. Phase 3 repeated decoder/no-mask controls at larger capacity through exp12–exp18, including short-horizon DiT-L and DiT-XL precursor runs used to validate stability before launching the final matrix. Phase 4 completed the DiT-XL 2×2 factorial: exp21 (baseline), exp22 (OT only), exp23 (mask only), and exp01 (both).

The mask-schedule controls are summarized in the committed DiT-B control configs exp03, exp05, exp06, and exp07. They were used to select the default $\gamma_{\max} = 0.75$ setting before the final DiT-XL runs; these short controls are stability checks rather than a final-scale FID sensitivity study. Values near 0.9 removed too many visible tokens near the noise-dominated end of the path ($t \approx 0$), making the masked velocity-regression problem unstable in short runs, while values near 0.5 reduced the compute benefit. The top-left exp01 YAML keeps `num_epochs=800` as a resume-friendly upper bound on PSC, but the factorial analysis uses the epoch-400 checkpoint so that all four cells align on the same comparison horizon.

All runs use PyTorch DDP with mixed precision, checkpoint-based resume, and Weights & Biases logging on PSC Bridges-2 NVIDIA H100-80GB GPUs. Training metrics, learning rate, global step, qualitative samples, and hyperparameter configurations are retained in W&B and form the evidence base for Tables 2 and 3. Each configuration in both tables is trained with a single random seed because of compute limits, so reported losses and FID values are point estimates rather than variance-bounded effect sizes.

4 Baseline and Extensions

4.1 Baseline Selection and Evaluation

Our primary baseline is an asymmetric latent DiT-XL flow-matching model trained without OT-CFM pairing and without adaptive masking. This corresponds to the A_{00} cell of the factorial design in §5.2, concretely registered as `configs/ablations/exp21.yaml`. We additionally report a U-Net reference trained with OT pairing only, to provide a second reference point at a different backbone family. Baselines are evaluated systematically on the same ImageNet-100 split, the same VAE latents, and the same optimizer schedule as the extended models, so that the primary comparison controls the data, architecture, optimizer, and offline evaluation protocol. The accepted mask-only cell uses a smaller global batch size than the other DiT-XL factorial cells, which we mark explicitly in Table 3 and treat as a limitation when interpreting isolated masking effects.

4.2 Implemented Extensions / Experiments

The two extensions on top of the DiT-XL flow-matching baseline are:

E1: OT-CFM. Minibatch Hungarian noise–data pairing (§3.1.2).

E2: Time-adaptive MaskDiT masking ($\gamma_{\max} = 0.75$). Cosine-squared token dropout during training only (§3.1.3).

E1 is intended to straighten training trajectories and reduce the number of function evaluations needed at inference time; E2 is intended to reduce encoder-side token processing during training. These mechanisms operate at different levels of the training loop, but their effects are not guaranteed to be additive, motivating the factorial ablation below.

4.3 Baseline Reproduction Evidence

The final baseline is exp21, trained for 400 epochs through the same checkpoint/resume and W&B pipeline as the other DiT runs. A shorter precursor, exp17, validated the XL no-mask setting before the full launch. Table 3 reports the reproduced baseline metrics, and the A_{00} / exp21 row of Figure 3 shows original samples from the same 400-epoch checkpoint: minimum training loss 0.6126 and $\text{FID}_{5k} = 15.2074$ under the shared Heun-40 / CFG-1.75 protocol.

5 Results and Analysis

We report training progress from our initial scale sweep, in which both OT pairing and adaptive masking are enabled simultaneously at DiT-S, DiT-L, and DiT-XL, plus a U-Net reference with OT pairing alone. Table 2 summarizes the training state of each configuration.

Exp	Backbone	OT	Mask	Steps	$\mathcal{L}_{\min}$
exp01	DiT-XL	✓	✓	290,990	0.624
exp02	DiT-L	✓	✓	141,492	0.690
exp04	DiT-S	✓	✓	463,090	0.657
exp15	U-Net	✓	—	67,699	0.667

Table 2: Preliminary training progress. Steps and minimum losses are taken directly from the logged W&B summaries.

Table 2 is descriptive rather than causal: the runs differ in backbone scale (Table 1), per-GPU batch size, and total steps reached before the 12-hour SLURM walltime. We use it only to show that the combined OT + mask method reaches a usable training regime across scales; causal attribution is reserved for the fixed-scale 2×2 factorial in §5.2.

5.1 Training Progress and Qualitative Samples

Table 2 reports training behavior through $\mathcal{L}_{\min}$, whereas Table 3 and Figure 2 report the standardized offline evaluation through FID_{5k} . The training-loss trajectory of exp01 (DiT-XL with both techniques) reaches a minimum of 0.624 in the preliminary sweep. exp15 (U-Net with OT only) serves as a reference point for loss-level comparison at a different backbone family. We retain the FID_{5k} point estimates as the primary outcome metric and use the qualitative panel only as visual corroboration. Figure 3 adds sample-level visual corroboration under the same Heun-40 / CFG-1.75 inference protocol.

5.2 Sensitivity / Ablation Analysis

5.2.1 2×2 Factorial Ablation

The central empirical question of this work — whether OT pairing and adaptive masking interact additively — requires a factorial ablation at a single fixed scale and comparable training horizon. We fix DiT-XL with a 400-epoch per-cell horizon (Table 3). The full-method cell, exp01, uses its epoch-400 checkpoint. The baseline exp21, OT-only exp22, and mask-only exp23 cells were evaluated with the same 400-epoch horizon and offline protocol.

Cell	OT	Mask	Global batch	$\mathcal{L}_{\min}$	FID_{5k}
A_{00} / exp21	—	—	128	0.613	15.2074
A_{10} / exp22	✓	—	128	0.582	15.7288
A_{01} / exp23	—	✓	64	0.605	15.4330
A_{11} / exp01	✓	✓	128	0.624	14.0405

Table 3: 2×2 factorial ablation at DiT-XL. All cells use a 400-epoch comparison horizon and Heun-40 / CFG-1.75 offline evaluation. The accepted mask-only run uses global batch size 64, so isolated masking effects should be read with that caveat.

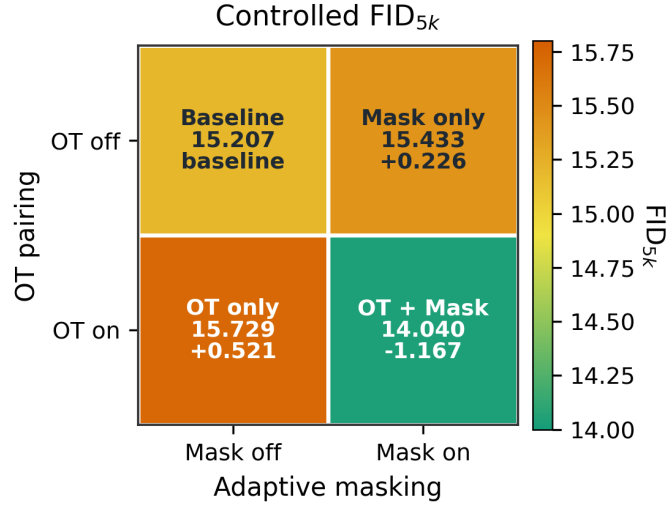


Figure 2: Controlled FID visualization for the fixed-scale DiT-XL factorial. Each cell reports FID_{5k} and the difference from baseline under Heun-40 / CFG-1.75; lower is better.

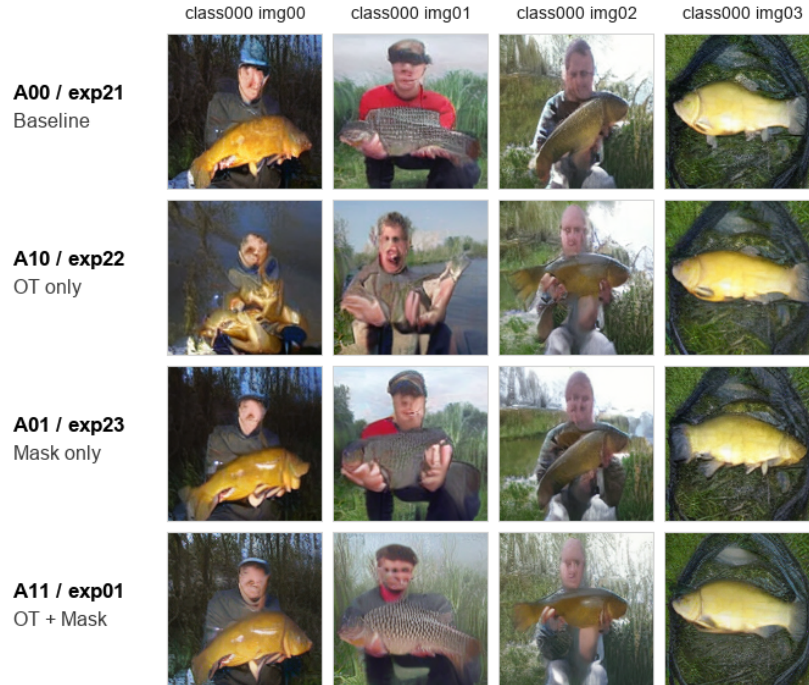


Figure 3: Qualitative sanity-check samples from the four DiT-XL factorial cells for a fixed ImageNet-100 class and fixed sample indices. Rows correspond to the cells in Table 3. The grid is used to inspect visible artifacts, while quantitative ranking is based on standardized FID_{5k}.

5.2.2 Scale Sweep as Secondary Ablation

The preliminary sweep in Table 2 is a secondary ablation along model capacity, with OT and mask jointly enabled. Since it lacks matched no-OT and no-mask controls at each scale, we read it as stability evidence only; causal attribution comes from the completed DiT-XL factorial in Table 3 and Figure 2.

5.2.3 Interaction-Effect Analysis Framework

Let y_{ij} denote the outcome (FID or $\mathcal{L}_{\min}$) for the cell with OT = i and mask = j . For FID_{5k}, the unscaled interaction is $y_{11} - y_{10} - y_{01} + y_{00} = -1.9139$; under the half-contrast convention this is $\hat{\alpha}_{\text{OT} \times \text{mask}} = -0.957$. Because lower FID is better, the negative interaction means that the combined cell beats the additive prediction. In direct terms, isolated OT and isolated masking worsen FID, while the combined cell improves over the baseline by 1.1669 points. These are single-seed point estimates with the mask-only batch-size caveat, so we interpret them as evidence of a compositional effect rather than a resolved effect size.

5.3 Error / Failure Case Analysis

One failure pattern appears in the preliminary scale sweep: minimum training loss does *not* decrease monotonically with model capacity. The DiT-L run (exp02, $\mathcal{L}_{\min} = 0.690$) is worse than both DiT-S (exp04, 0.657) and DiT-XL (exp01, 0.624), but it also reached fewer steps before walltime: 141 k versus 290 k for exp01 and 463 k for exp04. We therefore treat the scale sweep as stability evidence rather than a causal model-size comparison.

The more important final-stage failure mode is that the isolated interventions do not help under the fixed FID protocol: OT-only has the lowest training loss but the worst FID, and mask-only also trails the baseline. The positive result is therefore not an isolated OT or masking effect, but the interaction between noise–data pairing and token visibility in the combined cell.

Qualitatively, Figure 3 still shows visible image-level errors: fish textures are sometimes over-smoothed or distorted, human faces can lose detail, and foreground objects can blur into the background. These artifacts are consistent with the moderate FID_{5k} regime and motivate treating the sample grid as a sanity check rather than as the ranking metric.

6 Discussion

The central question is whether OT-CFM trajectory regularization and adaptive masking remain compatible inside one DiT training loop. Figure 2 shows that the full method beats the additive prediction from the isolated OT-only and mask-only cells. Since OT is computed before masking but gradients pass through the masked encoder, the matrix tests their interaction rather than treating them as independent improvements. Isolated OT/masking do not improve FID, and OT-only has the lowest training loss but worst FID; we therefore use FID_{5k} as the primary outcome.

Validity and limitations. The scale sweep verifies stability but is not causal: it uses one seed, unequal step counts, walltime truncation, and loss as a proxy. Final claims rely on the fixed-scale DiT-XL offline FID_{5k} matrix, but remain single-seed; exp23 uses global batch size 64 while the other cells use 128, and all results are on ImageNet-100 at 128×128. Future work should add seeds, rerun a batch-matched mask-only cell, extend benchmarks, and test alternative mask schedules.

7 Conclusion, GitHub, and Contributions

We presented minibatch OT-CFM pairing and time-adaptive MaskDiT-style token masking for class-conditional latent flow matching on ImageNet-100. The completed DiT-XL 2 × 2 matrix shows that the combined method gives the best FID_{5k}; code is at <https://github.com/MahikaGunjkar/Hw5DiffusionTeam6>.

Zhixuan Jiang handled latent modeling, DiT / flow-matching, OT/masking ablations, evaluation, and writing. **Mahika Gunjkar** and **Mahir Nagersheth** handled DDPM/DDIM baselines, class-conditional setup, the data pipeline, sampling scripts, and baseline experiments.

References

- J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255. IEEE, 2009. doi: 10.1109/CVPR.2009.5206848. URL <https://doi.org/10.1109/CVPR.2009.5206848>.
- S. Gao, P. Zhou, M.-M. Cheng, and S. Yan. Masked diffusion transformer is a strong image synthesizer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 23164–23173, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Gao_Masked_Diffusion_Transformer_is_a_Strong_Image_Synthesizer_ICCV_2023_paper.html.
- M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 6626–6637, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/8a1d694707eb0fefe65871369074926d-Abstract.html>.
- J. Ho and T. Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022. URL <https://arxiv.org/abs/2207.12598>.
- J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>.
- Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=PqvMRDCJT9t>.
- X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=XVjTT1nw5z>.
- I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- W. Peebles and S. Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Peebles_Scalable_Diffusion_Models_with_Transformers_ICCV_2023_paper.html.
- A.-A. Pooladian, H. Ben-Hamu, C. Domingo-Enrich, B. Amos, Y. Lipman, and R. T. Q. Chen. Multisample flow matching: Straightening flows with minibatch couplings. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 28100–28127, 2023. URL <https://proceedings.mlr.press/v202/pooladian23a.html>.
- R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.
- Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://openreview.net/forum?id=PxTIG12RRHS>.
- A. Tong, K. Fatras, N. Malkin, G. Hugué, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024. URL <https://openreview.net/forum?id=CD9Snc73AW>.
- H. Zheng, W. Nie, A. Vahdat, and A. Anandkumar. Fast training of diffusion models with masked transformers. *Transactions on Machine Learning Research*, 2024. URL <https://openreview.net/forum?id=vTBJtGioE>.